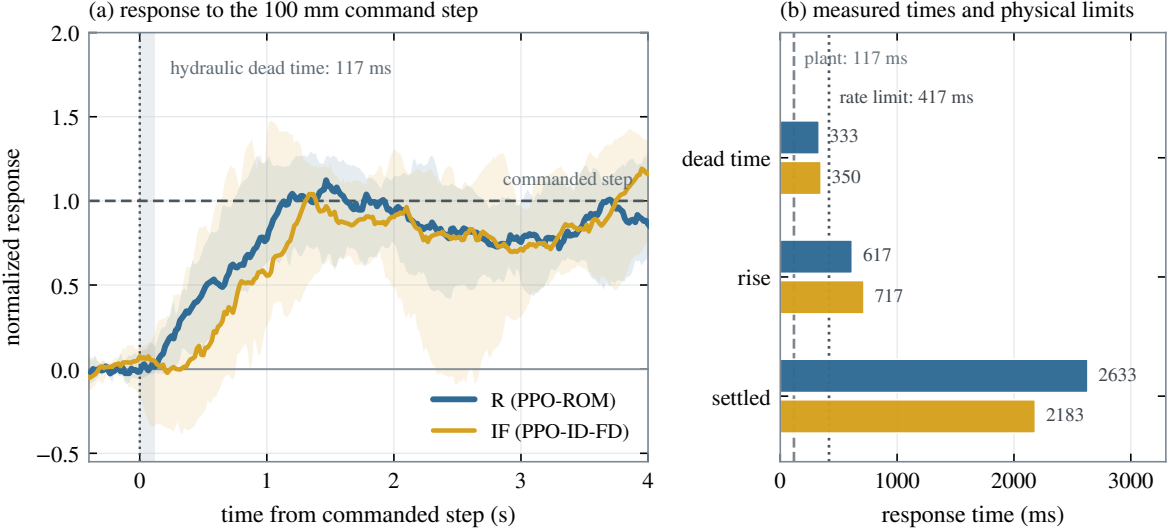
(a) response to the 100 mm command step
hydraulic dead time: 117 ms
commanded step
R (PPO-ROM)
IF (PPO-ID-FD)
normalized response
time from commanded step (s)
2.0
1.5
1.0
0.5
0.0
−0.5
0
1
2
3
4
(b) measured times and physical limits
plant: 117 ms
rate limit: 417 ms
dead time
rise
settled
333
350
617
717
2633
2183
0
1000
2000
3000
response time (ms)